

Exploratory Data Analysis of California Traffic Crash Data

Emily Miller

2026-02-04

Setup

```
library(tidyverse)
library(patchwork)
```

```
crash_raw <- read_csv(here::here("data", "hq1d-p-app52dopendataexport2025crashes.csv"))
party_raw <- read_csv(here::here("data", "hq1d-p-app52dopendataexport2025parties.csv"))
```

Data Wrangling

```
# Filter to at-fault drivers only for clean crash-level sobriety classification
# This avoids double-counting crashes with multiple drivers
party_drivers <- party_raw %>%
  filter(PartyType == "Driver",
         !IsOnDutyEmergencyVehicle,
         IsAtFault == TRUE) %>%
  select(CollisionId, SpeedLimit, SobrietyDrugPhysicalCode1,
         GenderCode, StatedAge) %>%
  janitor::clean_names()
```

```
# Join crash data with at-fault driver info and create derived variables
crash_clean <- crash_raw %>%
  select(`Collision Id`, `Crash Date Time`, `County Code`, NumberInjured,
         NumberKilled, IsFreeway) %>%
  janitor::clean_names() %>%
```

```

left_join(party_drivers, by = "collision_id") %>%
mutate(
  # Temporal variables
  date = mdy_hms(crash_date_time),
  hour = hour(date),
  month = month(date, label = TRUE, abbr = TRUE),

  # Outcome variables
  fatal = number_killed > 0,

  # Sobriety of at-fault driver: A = sober, B = impaired
  # Other codes are ambiguous so set to NA
  sobriety = case_when(
    sobriety_drug_physical_code1 == "A" ~ "No alcohol",
    sobriety_drug_physical_code1 == "B" ~ "Alcohol impaired",
    TRUE ~ NA_character_
  ),

  # Age bins for demographic analysis
  age_group = case_when(
    stated_age < 21 ~ "<21",
    stated_age < 30 ~ "21-29",
    stated_age < 45 ~ "30-44",
    stated_age < 65 ~ "45-64",
    TRUE ~ "65+"
  ),

  # Speed limit bins as proxy for road type
  speed_bin = case_when(
    speed_limit <= 30 ~ "0-30",
    speed_limit <= 45 ~ "35-45",
    speed_limit <= 60 ~ "50-60",
    TRUE ~ "65+"
  )
)

```

```

# Check how much data is lost by filtering to known sobriety codes
crash_clean %>%
  group_by(sobriety) %>%
  summarize(crashes = n_distinct(collision_id)) %>%
  mutate(prop = crashes / sum(crashes))

```

```
# A tibble: 3 x 3
  sobriety      crashes  prop
  <chr>         <int>  <dbl>
1 Alcohol impaired    29120 0.0774
2 No alcohol          213687 0.568
3 <NA>                133372 0.355
```

Temporal Patterns

```
# Compare hourly patterns: both groups vs alcohol-only view
tod_both <- crash_clean %>%
  filter(!is.na(sobriety)) %>%
  group_by(hour, sobriety) %>%
  summarize(crashes = n_distinct(collision_id), .groups = "drop") %>%
  ggplot(aes(hour, crashes, color = sobriety)) +
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = seq(0, 23, 3)) +
  labs(x = "Hour of day", y = "Total crashes", color = "Sobriety")

tod_alc <- crash_clean %>%
  filter(sobriety == "Alcohol impaired") %>%
  group_by(hour) %>%
  summarize(crashes = n_distinct(collision_id), .groups = "drop") %>%
  ggplot(aes(hour, crashes, color = "Alcohol Impaired")) +
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = seq(0, 23, 3)) +
  labs(x = "Hour of day", y = "Total crashes")

tod_both / tod_alc
```

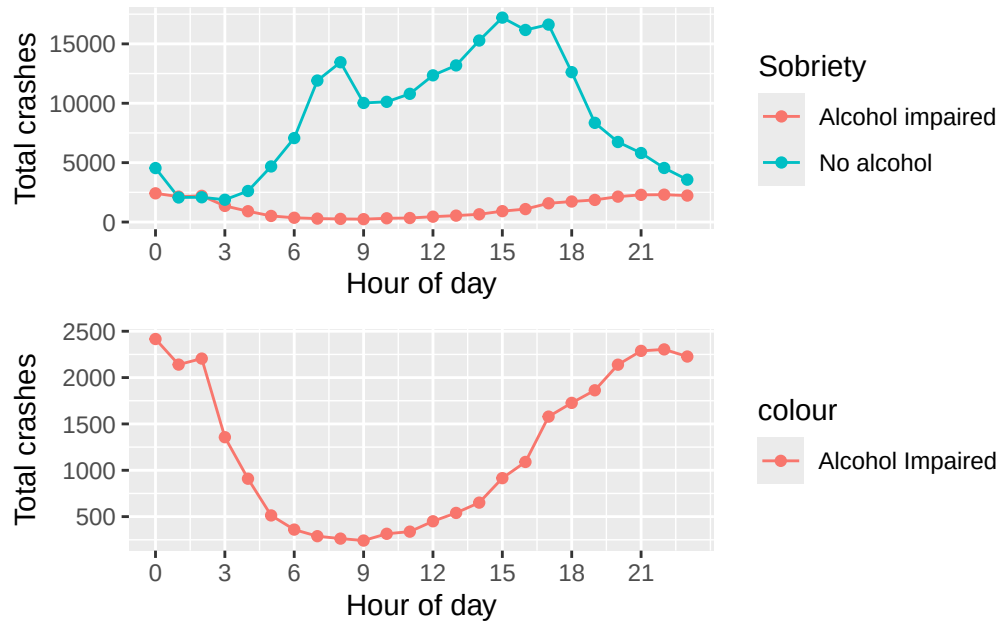


Figure 1: Crashes by hour

```
# Compare monthly patterns: both groups vs alcohol-only view
mo_both <- crash_clean %>%
  filter(!is.na(sobriety)) %>%
  group_by(month, sobriety) %>%
  summarize(crashes = n_distinct(collision_id), .groups = "drop") %>%
  ggplot(aes(month, crashes, color = sobriety, group = sobriety)) +
  geom_line() +
  geom_point() +
  labs(x = "Month", y = "Number of crashes", color = "Sobriety")

mo_alc <- crash_clean %>%
  filter(sobriety == "Alcohol impaired") %>%
  group_by(month) %>%
  summarize(crashes = n_distinct(collision_id), .groups = "drop") %>%
  ggplot(aes(month, crashes, group = 1, color = "Alcohol Impaired")) +
  geom_line() +
  geom_point() +
  labs(x = "Month", y = "Number of crashes")

mo_both / mo_alc
```

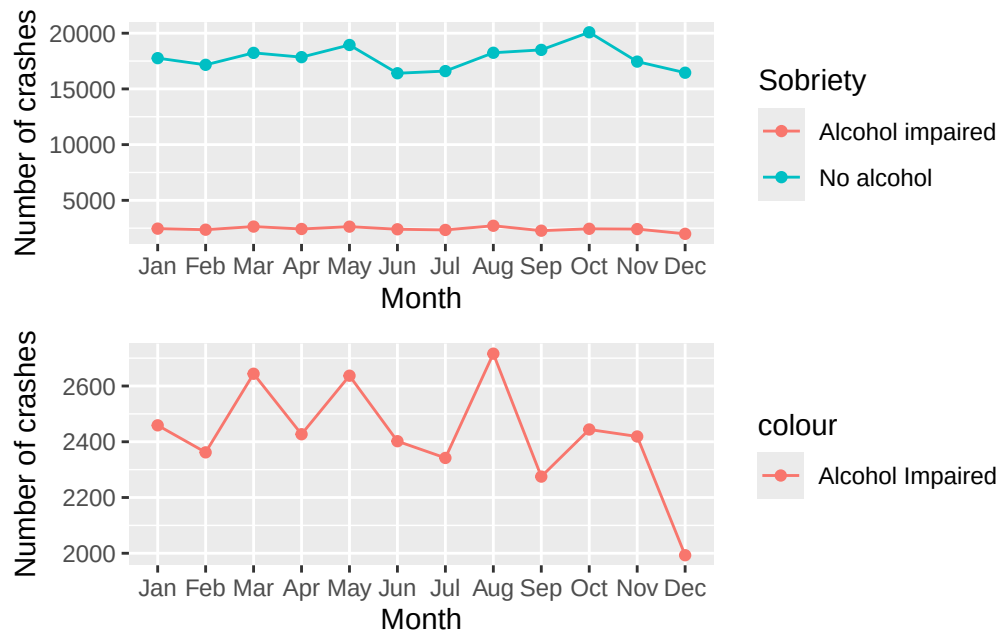


Figure 2: Crashes by month

Speed Limit / Road Type

```
# Crash counts by speed limit bin for both sobriety groups
crash_clean %>%
  filter(!is.na(sobriety), !is.na(speed_bin)) %>%
  group_by(speed_bin, sobriety) %>%
  summarize(crashes = n_distinct(collision_id), .groups = "drop") %>%
  ggplot(aes(speed_bin, crashes, fill = sobriety)) +
  geom_col(position = "dodge") +
  labs(x = "Posted speed limit", y = "Number of crashes", fill = "Sobriety")
```

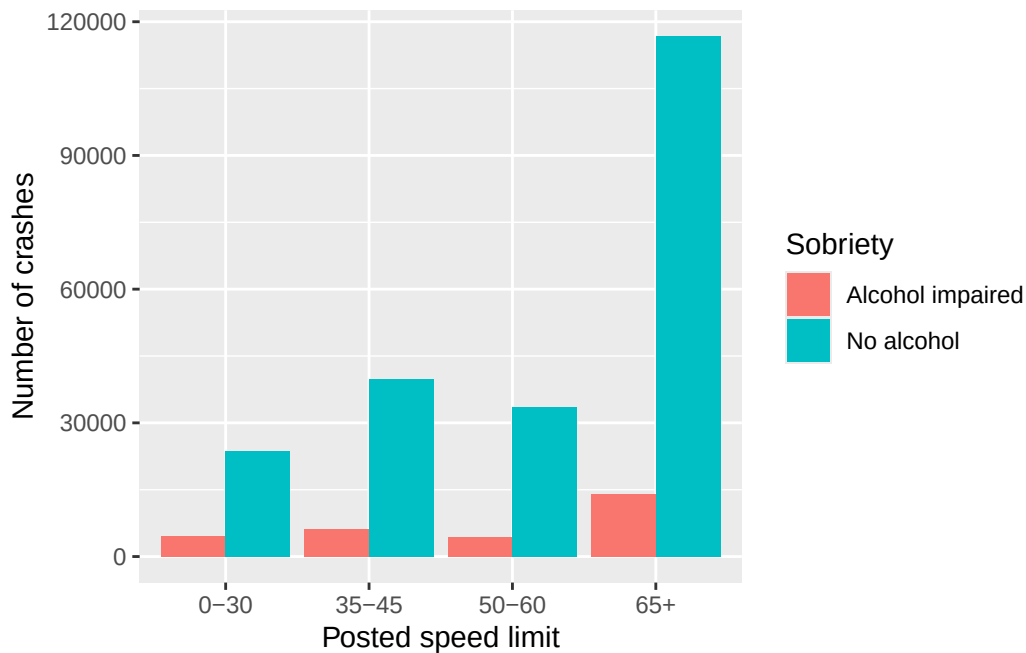


Figure 3: Crashes by speed limit

```
# Fatality rate by speed - shows severity rather than frequency
crash_clean %>%
  filter(!is.na(sobriety), !is.na(speed_bin)) %>%
  distinct(collision_id, sobriety, speed_bin, fatal) %>%
  group_by(speed_bin, sobriety) %>%
  summarize(
    crashes = n(),
    fatality_rate = mean(fatal, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  ggplot(aes(speed_bin, fatality_rate, fill = sobriety)) +
  geom_col(position = "dodge") +
  labs(x = "Posted speed limit", y = "Fatality rate", fill = "Sobriety")
```

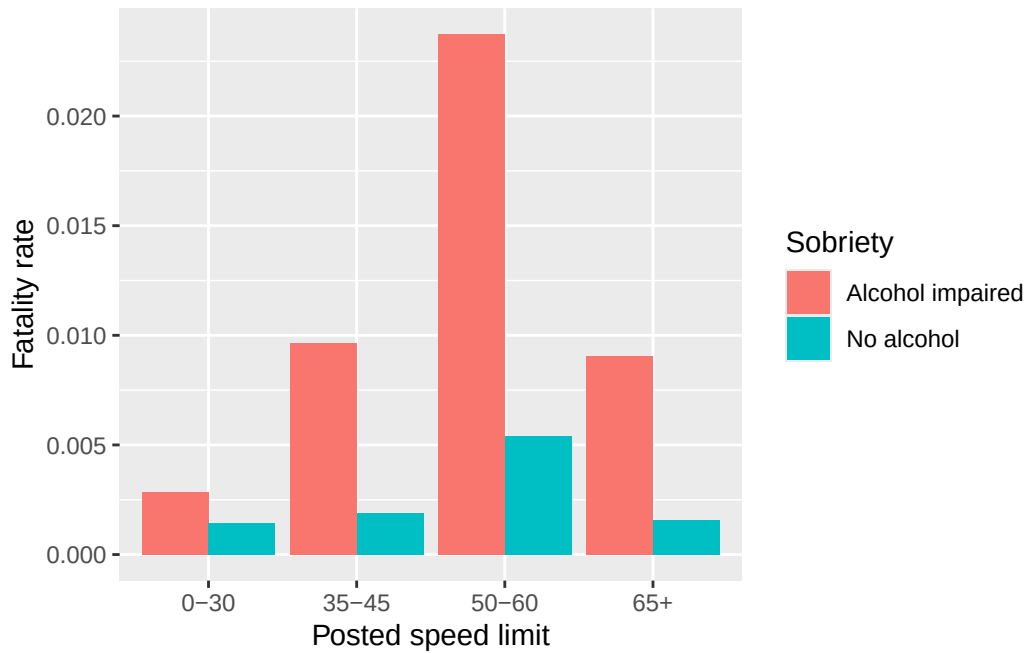


Figure 4: Fatality rate by speed limit

Demographics

```
# Age distribution of at-fault drivers by sobriety
crash_clean %>%
  filter(!is.na(sobriety), !is.na(age_group)) %>%
  group_by(age_group, sobriety) %>%
  summarize(crashes = n_distinct(collision_id), .groups = "drop") %>%
  ggplot(aes(age_group, crashes, fill = sobriety)) +
  geom_col(position = "dodge") +
  labs(x = "Driver age group", y = "Number of crashes", fill = "Sobriety")
```

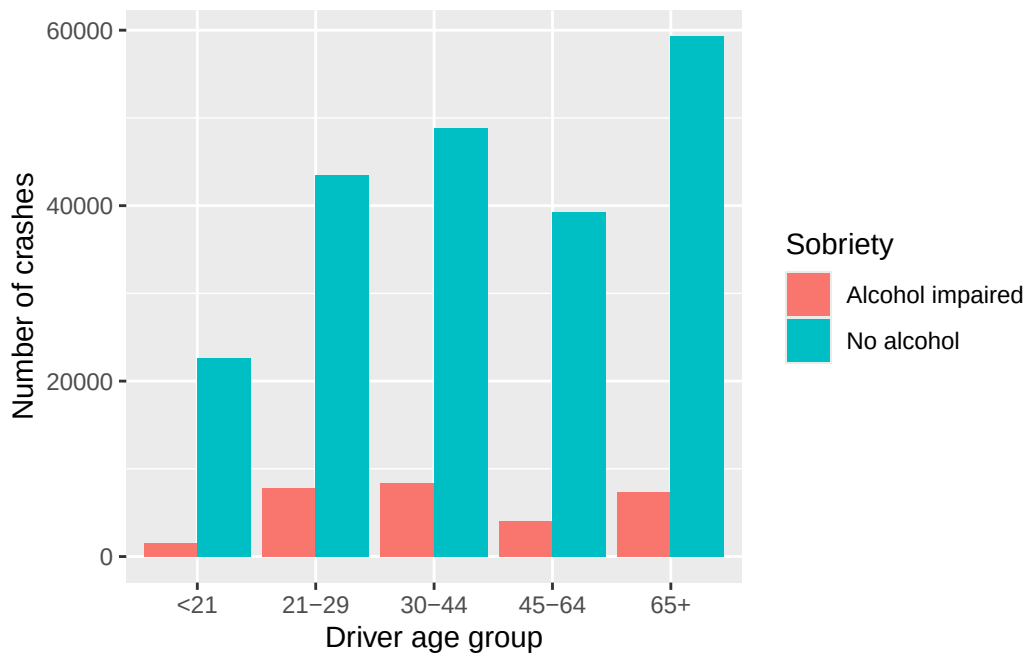


Figure 5: Crashes by age group

```
# Gender distribution - limited to M/F for cleaner comparison
crash_clean %>%
  filter(!is.na(sobriety), gender_code %in% c("F", "M")) %>%
  group_by(gender_code, sobriety) %>%
  summarize(crashes = n_distinct(collision_id), .groups = "drop") %>%
  ggplot(aes(gender_code, crashes, fill = sobriety)) +
  geom_col(position = "dodge") +
  labs(x = "Driver gender", y = "Number of crashes", fill = "Sobriety")
```

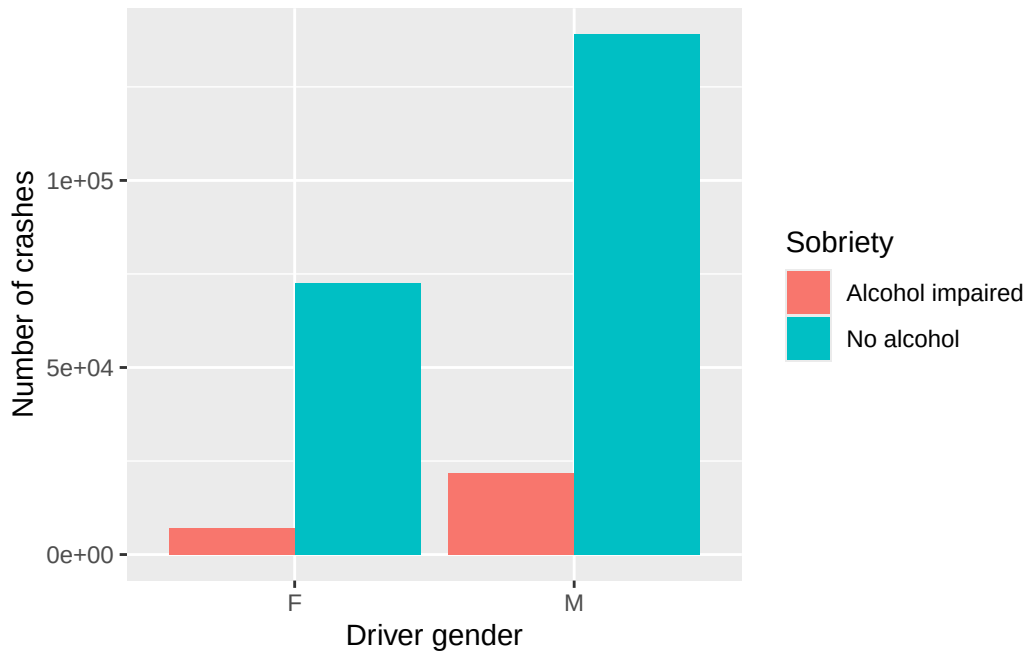



Figure 6: Crashes by gender

```
# Combined age and gender breakdown
crash_clean %>%
  filter(!is.na(sobriety), !is.na(age_group), gender_code %in% c("F", "M")) %>%
  group_by(age_group, gender_code, sobriety) %>%
  summarize(crashes = n_distinct(collision_id), .groups = "drop") %>%
  ggplot(aes(age_group, crashes, fill = sobriety)) +
  geom_col(position = "dodge") +
  facet_wrap(~ gender_code) +
  labs(x = "Driver age group", y = "Number of crashes", fill = "Sobriety")
```

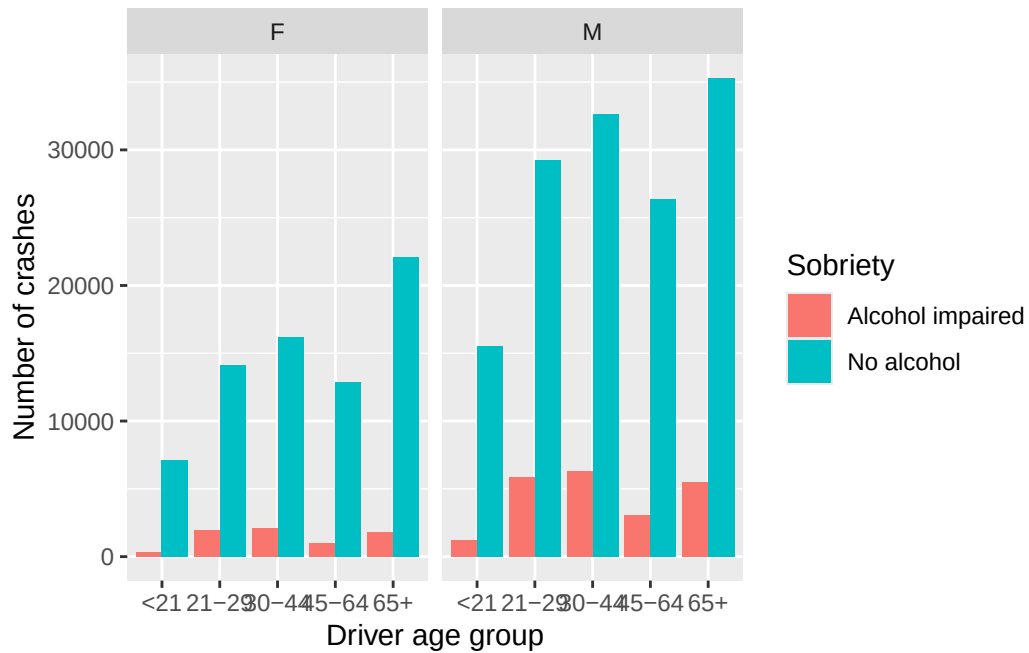


Figure 7: Crashes by age and gender

Overall Fatality Comparison

```
# Core comparison: overall fatality rate by sobriety status
crash_clean %>%
  filter(!is.na(sobriety)) %>%
  distinct(collision_id, sobriety, fatal) %>%
  group_by(sobriety) %>%
  summarize(
    crashes = n(),
    fatality_rate = mean(fatal, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  ggplot(aes(sobriety, fatality_rate, fill = sobriety)) +
  geom_col() +
  labs(x = "Driver sobriety", y = "Fatality rate")
```

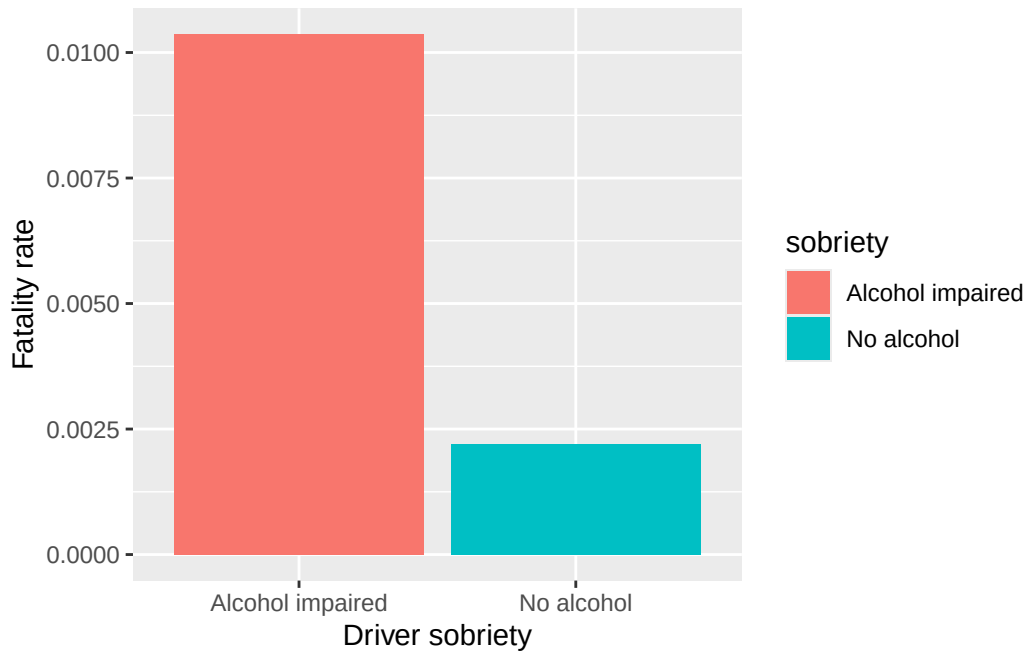


Figure 8: Overall fatality rate by sobriety

Part V Responses

1. What have you learned about your data? Have any potentially interesting patterns emerged?

Sober crashes are most common during daytime hours while alcohol-impaired crashes concentrate at night, as shown in Figure 1. The comparison panel reveals that when viewed in isolation, alcohol-impaired crashes appear confined to late-night hours. Sober crashes dominate overall volume. Seasonal patterns in Figure 2 are similar for both groups, with slight summer peaks.

Crashes for both groups occur on similar road types (Figure 3), with the number increasing with speed for both groups. The key difference emerges in Figure 4: fatality rates peak around 50-60 mph for both groups, but alcohol-impaired crashes are consistently more fatal at every speed level.

For demographics, Figure 5 shows older and middle-aged drivers dominate the sober group while those in their twenties and thirties are more prevalent in the alcohol-impaired group. Figure 6 shows men account for roughly twice as many crashes as women in both groups. These patterns persist across the combined view in Figure 7.

Overall, Figure 8 confirms that alcohol-impaired crashes are less common but more severe, with roughly double the fatality rate of sober crashes.

2. Have you made any strides towards answering your original questions?

Yes. The exploration shows that alcohol-impaired crashes can appear much less dangerous only the number of crashes are compared next to those with sober drivers. Plots focusing only on alcohol-impaired crashes by time (Figure 1 bottom panel) or month (Figure 2 bottom panel) suggest risk is confined to specific periods, which aligns with common but incomplete narratives ripe for manipulation.

Adding comparison groups and shifting from counts to rates changes that picture. Many apparent patterns in time, speed, and demographics largely reflect overall driving behavior, while severity measures like Figure 4 and Figure 8 consistently show higher risk when alcohol is involved across all conditions.

Next steps are to finalize which plot pairs best illustrate misleading versus honest framings, and experiment with axis truncation, selective time windows, and misleading annotations to see what kinds of plausible misinformation can emerge from these same data.

3. What challenges do you foresee?

The main challenge is making the misleading visualizations feel realistic rather than obviously flawed. Simple choices like using counts instead of rates are easy to spot, so the challenge is designing plots that look reasonable while still leading to incomplete conclusions. This requires careful choices about scale, aggregation, and framing.

There are also data limitations. A large share of records have unknown sobriety status, which limits the view of the full picture, and there is no exposure data such as miles driven by sobriety group. This means the analysis can compare crashes and severity, but not true risk per mile driven.